Cosmology with peculiar velocities of galaxies in the nearby Universe

I. Proposal's context, positioning and objective(s)

a. Objectives and research hypotheses

The accelerated nature of the expansion of our Universe remains one of the most mysterious puzzles in Physics, well consolidated by several recent and independent cosmological observations [1–5]. The expansion rate of the Universe $H(z)$ and the growth rate of the structures $f(z)$, both functions of cosmic time or redshift z, are two important observables that allow us to constrain dark energy models or alternatives to general relativity as potential solutions [6].The simplest model of *dark energy* is a cosmological constant Λ, which can also be thought as an exotic fluid with an equation of state $w = -1$. More exotic dark energy models predict different values for w, or that it can evolve with redshift. Recent cosmological observations suggest that $w \neq -1$ at more than 3σ, pointing to an time-evolving dark energy [5]. Alternatively to dark energy, some models attempt to extend or modify general relativity to account for the acceleration, the so-called *modified gravity theories*. To distinguish between dark energy and modified gravity models, it is essential to combine measurements of both expansion and growth rates [7]. **COSMOPV will constrain the dark energy equation of state as well as potential deviations from general relativity by measuring the growth rate of structures.**

The growth rate $f(z)$ quantifies how fast initial perturbations of the cosmic density field, $\delta(\vec{x})$, grow with time or redshift z. It can be measured via the impact of *peculiar velocities* of galaxies on the observed structures. Peculiar velocities modify redshifts due to the cosmic expansion. Both are generally undistiguishable, creating distortions in the observed galaxy density field, known as *redshift-space distortions* (RSD). The two-point statistics of the galaxy density field, $\langle\delta(\vec{x})\delta(\vec{y})\rangle$, becomes anisotropic w.r.t. the line-of-sight, allowing measurements of the growth rate $f(z)$ with spectroscopic galaxy surveys [8, 9]. Moreover, if distance indicators are available for a large sample of galaxies, it is possible to break the degeneracy between expansion and proper motion, yielding a radial velocity field $v(\vec{x})$. Several distance indicators have been used in recent years: the Tully-Fisher (TF) and Fundamental Plane (FP),relations, as well as type-Ia supernovae (SNIa),The first two indicators are abundant but yield more uncertain distances (about 20-30%) than the last one (5-10%), though less numerous. The two-point statistics of the velocity field, $\langle v(\vec{x})v(\vec{y})\rangle$, also depends on $f(z)\sigma_8(z)$. Measuring three combinations of two-point functions, i.e., density-density, velocity-velocity and their cross correlation $\langle\delta(\vec{x})v(\vec{y})\rangle$, yield tighter constraints than just one of such statistics [10–13]. **COSMOPV will employ state-of-the-art samples of distance indicators (FP, TF, and SNIa) to measure peculiar velocities and obtain the tightest constraints on the growth rate at late times.**

b. Position of the project in relation to the state of the art

The COSMOPV team has more than five years of expertise in this field and it is currently well established as leads in key cosmological analyses with peculiar velocities, including surveys such as:

- **the Dark Energy Spectroscopic Instrument (DESI):** currently the best spectroscopic survey of galaxies to date, it will measure 63 million galaxy redshifts after eight years of observations (2021-2028) over 17,000 deg^2 [14]. DESI galaxies are an excellent tracer of the distorted density field, from which we can measure anisotropies due to RSD. The DESI Peculiar Velocity survey will produce distance measurements to 53k spiral galaxies from the Tully-Fisher relation and 133k ellipticals from the Fundamental Plane [15],Those distances will yield a well sampled velocity field at $z < 0.1$.
- **the Zwicky Transient Facility (ZTF):** a photometric survey with large field of view and high cadence, ideal for the discovery of transient events, including SNIa [16]Covering more than 17k deg^2 of the Northen sky, perfectly overlapping with DESI observations, ZTF will discover more than 30k SNIa over redshifts $0 < z < 0.2$. which will be used to obtain precise peculiar velocities.

COSMOPV will focus in the DESI and ZTF overlap, a perfect opportunity for cross-correlations and systematic cross-checks. Based on our own Fisher forecasts [17, 18], **COSMOPV will produce a 8.3% precision growth**

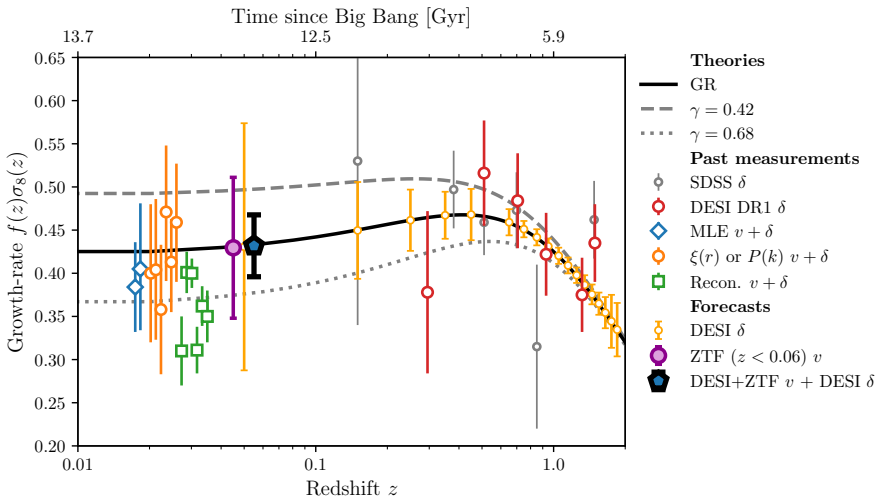

Figure 1. Growth rate of structures versus redshift from past measurements compared to expected results from COSMOPV, which will employ low-redshift galaxies and peculiar velocities from the Dark Energy Spectroscopic Instrument (DESI) and the Zwicky Transient Facility (ZTF).

rate measurement when combining DESI galaxies and DESI+ZTF velocities (Figure 1), representing the most precise and robust estimate at late cosmic times.

Producing large samples of synthetic datasets reproducing real data at high fidelity is an essential ingredient. Those sets have multiple uses: testing our methodologies for biases, understand our uncertainties, compute covariance matrices of observables, and study the impact of systematic effects on cosmological results. Building up on current experience in our team [17, 19–23], **COSMOPV will lead the production of the most realistic synthetic datasets for both DESI and ZTF.**

c. Methodology to reach the scientific objectives of the project

COSMOPV is organised in three connected work packages: synthetic datasets, systematic effects in real data, and cosmological implications of our results.

Work package 1: Synthetic datasets

WP1 will focus on the production and analyses of realistic synthetic datasets reproducing properties of real data from DESI and ZTF.

- **Fast approximate simulations** Recent mock catalogs used for peculiar velocities are based on n-body simulations such as AbacusSummit [23]or Uchuu [24], which are high accuracy but are limited in volume or in the number of independent realisations. Using well-known techniques [20, 25], we will produce a large number ($O(1000)$) of independent realisations of density and velocity fields at low redshift.
- **Production of synthetic DESI Peculiar velocity surveys and ZTF SNIa samples:** We will add survey-like properties to the simulations above to create sythetic samples of DESI and ZTF surveys, including distance indicators. For DESI, we will use existing infrastructure used to produce DESI Data Release 1 mock catalogues for TF and FP samples [23] while for ZTF, we will build upon in-house experience [17] and use existing software (SKYSURVEY) from the collaboration. Those new mock catalogues will reach higher redshifts than current sets, and will include subtle effects of redshift evolution and selection effects.
- **Cross-covariance between methodologies and cosmological probes:** Several methodologies are able to exploit a distorted density field and a peculiar velocity field (e.g. [13, 18]). Those are commonly applied to the same initial dataset, so results are correlated. Using large sets of mock catalogues, we will precisely estimate this cross-correlation, which is key to derive consensus results (e.g. [26–28]). The same mocks could be further improved to evaluate the cross-covariance between growth measurements from peculiar velocities and weak lensing.

Work package 2: Systematic effects in real data

Focusing on real data, WP2 will investigate potential systematic effects affecting growth measurements and will yield the final combined result from DESI and ZTF.

- **Quantify the impact of observational systematics:** With the mocks from WP1, we will quantify for the first time the impact of several observational systematics to an unprecedented level of accuracy. Several systematic effects can create spurious correlations in the observed velocity field, potentially biasing our growth rate measurements [29]. We will pay particular attention to the angular dependence of the Malmquist bias, dust attenuation, flux calibration residuals, among others. Strategies will be put in place to mitigate such impacts when possible, otherwise a systematic uncertainty budget will be propagated to final results.
- **Growth rate measurement with a photometric sample of SNIa:** Building up on our experience with synthetic datasets of Rubin LSST SNIa [22, 30], we will extend this work to the final release of ZTF SNIa data. This photometrically-classified sample can potentially increase the number of SNIa by factors of 5 to 6 compared to the spectroscopically-classified sample, also increasing the redshift range to $0 < z < 0.2$.
- **Cross-calibration of DESI and ZTF peculiar velocities:** Given the existence of three different distance indicators (TF, FP and SNIa) in the same volume, there are many opportunities for cross-calibrations and consistency checks between those, in regards to cosmological parameters. Cross-correlations between different surveys are much less sensitive to systematic effects, so measuring them adds to the robustness of our analysis. We aim at obtaining a final consensus growth rate measurement from this combination, including all contributions of potential systematic errors.

Work package 3: Cosmological implications of our measurements

WP3 will derive cosmological constraints from results of WP1 and WP2. Our growth rate measurement at low redshift, when combined with high redshift expansion and growth rate measurements (Fig. 1), has the potential to improve constraints on dark energy and rule out some modified gravity theories.

The expansion rate will be provided by SNIa Hubble diagrams, baryon acoustic oscillations and the cosmic microwave background. At high redshift, DESI and Euclid will also provide precise growth rates from RSD. With those probes, WP3 will deliver constraints on:

- Ω_m, σ_8, H_0 when assuming a ΛCDM model
- w_0 and w_a parameters describing the equation of state of evolving dark energy
- γ or (μ_0, Σ_0) which empirically parametrise deviations from general relativity. In-house forecasts predict that the uncertainty on γ reduces by a factor of ~ 2 when including our low-redshift growth rate measurement.

Those results will set the tightest constraints from the Northen sky. This work serve as basis for future analyses of datasets in the Southern sky such as Rubin LSST and 4MOST.

II. Organisation and implementation of the project

COSMOPV is a PRME project that will be hosted at Aix-Marseille Université (amU), particularly at the Centre de Physique des Particules de Marseille (CPPM). The team consists of researchers already active in the field, all members of either DESI and ZTF collaborations, spaning a wide range of expertise.

a. The scientific team at CPPM

- **Julian Bautista - Scientific Coordinator (PR, 90% ETPR):** Expert in galaxy clustering, type-Ia supernovae and peculiar velocities for cosmology. Currenly active member of DESI and ZTF and actively working in peculiar velocities for the past 5 years, with the support of the Chaire d'Excellence A*Midex. Current co-chair of the DESI Transients and Low-redshift Cosmology Working Group, whose main goal is to perform growth rate measurements with the DESI Peculiar Velocity Survey.

- **Dominique Fouchez (DR1, 10% ETPR):** Expert in supernova cosmology (SNLS, ZTF and LSST), photometric detection of sources as well as applications of machine learning to astronomy.
- **Benjamin Racine (CRCN, 10% ETPR):** Expert in cosmology with the CMB and SNIa. Leading the efforts to homogenise the flux calibration of ZTF images to produce datasets suited for precision cosmology with SNIa. His raw data expertise is of extreme value for COSMOPV, as he will be able to advice on potential sources of systematic effects that impact cosmological results.
- **Fabrice Feinstein (PR, 40% ETPR):**
  He has been closely working with B. Racine on ZTF datasets and has experience in supernova cosmology. Will contribute with the investigation of observational systematics in the data.
- **Juan Mena-Fernandez (Postdoc, 30% ETPR):**
  Expert in galaxy clustering, weak lensing and mock catalog production. Will be joining the team in Nov 2025 for three years and part of his work will be related to COSMOPV.
- **Corentin Hanser (Postdoc, 30% ETPR):**
  Transitioning from cosmology with galaxy clusters to peculiar velocities. Will contribute during one year to the project.
- **Rafael Kebadian (PhD student, 30% ETPR):**
  Currently leading the measurement of the growth rate of structures with half of ZTF data, will be part of the project during his last year.

b. Implemented and requested resources to reach the objectives

For the success of COSMOPV, we would like to request the funding of one doctoral researcher (137 977,00 €) and one experienced postdoctoral researcher (253 378,00 €), each for the duration of 36 months. Each position will be assisted with funds for a work computer (2k € each) and travel support (2.5k € per year per person and 1k € per year for the scientific coordinator). Our results will be published in the peer-reviewed journal Astronomy and Astrophysics, and we request 3k € for eventual associated publication fees. Including fees, **the total budget of COSMOPV amounts to 473 697,93 € for a duration of 48 months.**

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